Article

Am J Hosp Pharm 36:1321–1325 (Oct) 1979

Evaluation of a clinical pharmacist in caring for hypertensive and diabetic patients

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A randomized controlled trial was conducted in a medical follow-up clinic to assess the effects of intervention by clinical pharmacists on the management of hypertensive and diabetic patients.

Before and after the trial (March 1976 through August 1978), the chronic disease status of patients who received conventional care from a single physician (control group) was compared with the status of patients who received care directly from a clinical pharmacist (experimental group). The care provided to the experimental patients was monitored closely by physicians on the medical school faculty.

The outcome of care provided to patients in the control and experimental groups was equivalent, in terms of diastolic blood pressure and fasting blood sugar levels. However, the control group averaged significantly lower mean systolic pressure ($p \leq 0.02$). Patient satisfaction appeared greater in the experimental group, as judged by a statistically higher kept-clinic-appointment rate ($p \leq 0.0005$) and a lower dropout rate ($p \leq 0.02$). The rate of compliance with therapeutic regimens was higher in the experimental group, but not significantly higher. There were four deaths in the experimental group and 11 in the control group.

This study provides additional evidence to justify the safe and effective role of the clinical pharmacist in the postdiagnostic management of patients with diabetes mellitus and hypertension.

Key words: Antidiabetic agents; Clinical pharmacists; Errors, medication; Hypotensive agents; Patient information

The clinical roles pharmacists proclaim have been subjected to critical assessment only rarely. In fact, we found only five published studies in which a systematic evaluation of outcomes of care was made following clinical pharmacy intervention. In three of the studies,[1–3] the pharmacists principally were involved in monitoring drug therapy and providing rational drug information to physicians and patients. Some important benefits were demonstrated in the results of all three investigations. In the remaining two studies,[4,5] pharmacists were delegated the responsibility for taking care of selected patients, and the reported results indicated the pharmacists provided safe and effective care.

To evaluate more extensively the effectiveness of pharmacists in providing patient care, we conducted a 29-month randomized controlled trial of a clinical pharmacist in the postdiagnostic management of patients with hypertension and diabetes mellitus. It was our thesis that the outcomes of clinical pharmacy care would be equivalent to those resulting from physician care. Accordingly, the hypothesis that the management of patients by a pharmacist is safe and effective would be supported if no statistically and clinically significant difference could be shown between the outcomes of care of two comparable groups of patients: one group managed by a clinical pharmacist with physician review, the other group by a physician.

Methods

The medical follow-up clinic at the Robert B. Green Hospital in San Antonio, TX, was chosen as the study site. The hospital is operated by the Bexar County Hospital District and serves as a primary care training facility for students and residents of The University of Texas Health Science Center at San Antonio. More than 50 outpatient clinics meet regularly at the Robert B. Green Hospital, and more than 100,000 patients receive care in the clinics annually.

The medical follow-up clinic meets five days a week and is staffed by a physician who is an associate professor of

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The assistance of Linda Castro Barragan is acknowledged.

The study was supported by DHEW Public Health Service grant 06-000100-01.

0002-9289/79/1001-1321$01.50

family practice, and four licensed vocational nurses (LVNs). As of March 1, 1976, there were 1,722 patients enrolled in the clinic. More than 90% of these patients were Mexican-American and more than 95% were indigent. Although a few of these patients were known to have other chronic diseases such as hypothyroidism and congestive heart failure, the majority of the patients were being followed for hypertension, diabetes mellitus or both.

As a preliminary to this study, a joint agreement was made between the Department of Family Practice at the medical school, the College of Pharmacy at Austin and the Bexar County Hospital District to delegate direct patient care responsibility to a clinical pharmacist appointed to the medical follow-up clinic. A Pharm.D. with two years of clinical training in general medicine was chosen to fill this position. The Department of Family Practice agreed to monitor closely the patient care activities of the clinical pharmacist.

Patient Assignment. Because a case load of approximately 600 patients was considered manageable for a clinical pharmacist, the eligible patients were assigned randomly into three groups. Each group consisted of 574 patients; the first and second groups were designated as the experimental and control group, respectively. The pharmacist, assisted by doctor of pharmacy candidates, managed the experimental patients. The clinical physician, aided by four LVNs, provided care to the control patients. The third group, identified as a residual group, continued to receive care from the clinical physician, but data generated by this group were not included in the analysis of this study except to determine a kept-clinic-appointment rate for the control group, as discussed below.

Management of the experimental group's patients was monitored closely by physicians in the Department of Family Practice to assure the provision of adequate medical care. However, these physicians were not involved directly in the care of these patients, nor were they involved in any way in the care of the control group's patients.

Assessment Criteria. The criteria selected for evaluating the effects of pharmacist intervention on the follow-up care of hypertensive and diabetic patients included: (1) kept-clinic-appointment rate; (2) rate of compliance with antihypertensive therapeutic regimens; (3) frequency of follow-up clinic visits; (4) frequency of emergency room walk-in visits; (5) frequency of referral clinic visits; (6) frequency of hospital admissions; (7) systolic blood pressures; (8) diastolic blood pressures; and (9) fasting blood sugar levels.

The kept-appointment rate for the medical follow-up clinic was determined by the director of ambulatory care services for the hospital. His records indicated the number of follow-up clinic appointments made by the clinical pharmacist and the clinic physician, and the number of patients who failed to keep their appointments with either the pharmacist or physician. Thus, a kept-clinic-appointment rate for the experimental group was calculated readily from the available appointment data. However, since the appointment records did not separate control from residual group patients, we used the combined data from these two groups to estimate the kept-clinic-appointment rate of the control group.

To measure the rate of compliance with antihypertensive regimens, we randomly selected 100 hypertensive patients from both the experimental and control groups who, at the start of the study, were taking either a single diuretic drug or a combination of a diuretic drug plus methyldopa. Compliance rate was calculated using the formula:

$$\text{Compliance rate} = (\text{No. of doses dispensed}/\text{No. of doses prescribed})\ 100$$

The prescribing and dispensing information needed for this formula was extracted from each patient's medication profile record. Investigation of compliance was limited to the period from January 1977 to January 1978. The criterion for good compliance was set at equal to or greater than 80%, in accordance with the work done by Sackett et al[6] which showed that it was only above this level of compliance that blood pressure fell systematically. Only patients completing one year of continuous, unaltered therapy were included in the analysis.

To determine the frequency and type of health care interventions (criteria 3 through 6) and to assess the clinical outcomes of care (criteria 7 through 9), we audited the medical records of all patients who completed the trial. The number of outpatient visits and hospital admissions made by each patient during a two-year period (March 22, 1976 through March 22, 1978) were recorded and analyzed.

A pretrial-posttrial assessment of systolic and diastolic blood pressures and fasting blood sugar levels was used to evaluate the effect of clinical pharmacy intervention on the outcomes of patient care. These particular criteria were chosen because of their wide acceptance as objective indicators of control of hypertension and diabetes mellitus. For each outcome criterion, a set of six observations was used to calculate a mean assessment. The pretrial observations were made during a 12-month period immediately preceding the initiation of the study. The posttrial observations were made 24 to 29 months after the study began.

In addition to comparing the means of each assessment criterion, we also evaluated the number of patients in each group whose hypertensive or diabetic condition improved, deteriorated or remained unchanged during the course of the trial. To facilitate the analysis of changes in the patients' pretrial and posttrial clinical conditions, the blood pressure and fasting blood sugar data were stratified according to the scheme:

A. Systolic blood pressure
   1. ≤160 mm Hg
   2. >160 mm Hg
B. Diastolic blood pressure
   1. ≤95 mm Hg
   2. >95 mm Hg
C. Fasting blood sugar
   1. <150 mg/dl
   2. 150 to 199 mg/dl
   3. ≥200 mg/dl

This scheme was chosen arbitrarily by the physician audi-

Table 1. Reasons for Attrition in Experimental and Control Groups

Variable	Experimental Group No.	Experimental Group %	Control Group No.	Control Group %	Significance[a]
Randomly assigned patients	574	100	574	100	—
Patients failing to enter study	115	20.0	147	25.6	$p \le 0.02$ ($z = -2.26$)
Patient dropouts	84	14.6	111	19.3	$p \le 0.02$ ($z = -2.12$)
Exclusions	13	2.2	11	1.9	NS
Deaths	4	0.7	11	1.9	NS
Unlocated medical records	9	1.6	14	2.4	NS
Total attrition	225	39.2	294	51.2	$p \le 0.01$ ($z = -4.08$)
Medical records audited	349	—	280	—	

[a] NS = no significant difference.

tors to represent different levels of control of hypertension and diabetes mellitus; i.e., adequate or inadequate control of systolic or diastolic blood pressure (stratum 1 versus stratum 2), and good, fair or poor control of diabetes mellitus (stratum 1 versus stratum 2 versus stratum 3).

Statistical Analysis. The research involved the use of two experimental designs. Data for measuring the outcomes of hypertension and diabetes mellitus (criteria 7 through 9) were generated by a pretest-posttest control group design. A posttest-only control group design was used to obtain the data for assessing criteria 1 through 6.

Statistical analysis of the data was accomplished using chi square (incorporating Yate's correction for continuity where appropriate), a normal test (z-test) for proportions, and the t-test for independent samples. An analysis of covariance also was used to adjust posttest assessments for initial pretest differences.

Results

Of the 574 patients randomly assigned to each group, 349 (60.8%) experimental patients and 280 (48.8%) control patients completed the 29-month trial. Table 1 shows the control group experienced a significantly greater patient dropout rate and total attrition. A substantial number of patients in both groups never used the clinic while the study was in progress. Table 2 gives the reasons why some patients in both groups were excluded from the study.

Some of the characteristic features of the patients who completed the trial are given in Table 3. The groups were different in two respects. First, the control group contained a higher percentage of patients with hypertension as an only diagnosis. Secondly, a higher percentage of experimental patients had both hypertension and diabetes.

The kept-clinic-appointment rate and the rate of adherence to antihypertensive therapeutic regimens were used to assess patient compliance. The kept-clinic-appointment rate was significantly higher for the experimental group than that for the control and residual groups combined (Table 4). On the other hand, the difference in the percentages of good therapeutic compliers (>80% compliance rate) between a random sample of hypertensive patients selected from each group was not statistically significant for either single or multiple drug treatment (Table 5).

The impact of pharmacist intervention on the volume of outpatient clinic visits and the number of hospital admis-

Table 2. Reasons for Excluding Patients from Study

Reasons for Exclusions	No. of Patients Experimental	No. of Patients Control
Dismissed because of absence of chronic illness	4	5
Incomplete medical record	4	3
Enrolled in another study	3	2
Transferred to another health care facility	1	1
Late entry into study	1	0
Total	13	11

Table 3. Characteristics of Study Patients

Characteristic	Experimental (N = 349) No.	Experimental (N = 349) %	Control (N = 280) No.	Control (N = 280) %	Significance[a]
Mean age	61		60		NS
Percent female		75.6		77.5	NS
Hypertensive only	146	42	137	49	$p \le 0.04$ ($z = -1.779$)
Diabetic only	87	25	69	25	NS
Hypertensive and diabetic	99	28	60	21	$p \le 0.025$ ($z = -1.99$)
Neither hypertensive nor diabetic	17	5	14	5	NS

[a] NS = no significant difference.

Table 4. Kept-clinic-appointment Rates[a]

Group	Appointments Kept Yes	Appointments Kept No	Appointments Kept Total	%
Experimental	3,744	750	4,494	83.3
Control and residual	9,174	3,247	12,421	73.8
Total	12,918	3,997	16,915	76.3

[a] $z = 12.83$, $p \le 0.0005$.

Table 5. Compliance Evaluation of Hypertensive Patients

Medication	Compliance Rate	Experimental No.	Experimental %	Control No.	Control %	Total
Diuretic drug only[a]	≥80%	23	60.5	18	52.9	41
	<80%	15	39.5	16	47.1	31
Diuretic plus methyldopa[b]	≥80%	33	84.6	17	65.4	50
	<80%	6	15.4	9	34.6	15

[a] Chi square = 0.42; degree of freedom = 1; $p \le 0.7$.
[b] Chi square = 2.26; degree of freedom = 1; $p \le 0.2$.

sions is reflected by the data shown in Table 6. The care provided by the clinical pharmacist resulted in significantly more follow-up clinic visits. However, the differences in the frequency of other outpatient visits and the number of hospital admissions between the two groups were not statistically significant.

Table 7 contains the mean group clinical data used in assessing the outcomes of care in the hypertensive and diabetic patients. The *t*-test was used to determine the significance of the difference in the means of the posttest assessments between the two groups. Analysis of the data indicates a significant difference between the two groups only with respect to systolic blood pressure. At the end of the trial, patients in the control group had a significantly lower systolic blood pressure than did patients in the experimental group.

Since a significant difference between the two groups was found in the initial fasting blood sugar (FBS) assessments, the fasting blood sugar data also were subjected to analysis of covariance (Table 8). The application of this statistical method revealed a difference in the post-FBS assessments which fell short of achieving significance at the 0.05 level ($p < 0.058$).

In addition to testing the significance of the differences in the means of posttest assessments, we also evaluated the number of patients whose hypertensive or diabetic condition deteriorated or improved during the study. A chi-square test for independence was used to evaluate the data. Tables 9 through 11 summarize the statistics. Table 9 confirms a difference in outcomes between the two groups with regard to systolic blood pressure. The data in Table 10 could not be analyzed since chi-square requirements were not met by the data. Although Table 11 shows some differences in the outcomes of diabetic care between the two groups, these differences were not statistically significant.

Discussion

The results of this research indicate that care by the pharmacist was equivalent to care by the physician with respect to the rate of compliance achieved with antihypertensive drug regimens, and the degree of control obtained

Table 6. Frequency of Clinic Visits and Hospital Admissions

Type of Service	Mean No. Patients/Yr Experimental (N = 349)	Control (N = 280)	Significance[a]
Follow-up clinic visits	6.69	5.38	$t = 7.63$, df = 627, $p \le 0.001$
Referral clinic visits	2.63	2.84	NS
Emergency room walk-in visits	1.18	1.01	NS
Hospital admissions	0.16	0.17	NS

[a] df = degrees of freedom; NS = no significant difference.

Table 7. Comparison of the Mean Blood Pressures and Fasting Blood Sugars of Patients, Pretrial and Posttrial

Variable	Experimental Group		Control Group	
	Pre	Post	Pre	Post
Mean systolic blood pressure (mm Hg)[a]	145 (±15)[c]	147 (±18)	143 (±14)	141 (±13)
Mean diastolic blood pressure (mm Hg)	86 (±6)	84 (±6)	86 (±6)	84 (±4)
Mean fasting blood sugar (mg/dl)[b]	192 (±46)	184 (±42)	182 (±39)	189 (±49)

[a] Posttest systolic blood pressures are different: $p = \le 0.001$, $t = 3.88$.
[b] Pretest fasting blood sugars are different: $p \le 0.05$, $t = 1.98$, degrees of freedom = 313.
[c] Standard deviation.

Table 8. Analysis of Covariance—Posttest, Fasting Blood Sugar (FBS)

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F Statistic	Significance of F
Covariate pretest FBS	116,916.000	1	116,916.000	69.952	0.001
Main effects group	6,029.625	1	6,029.625	3.608	0.058
Residual	521,466.250	312	1,671.366	—	—
Total	644,411.875	314	2,052.267	—	—

Table 9. Number of Hypertensive Patients Whose Systolic Blood Pressure Increased, Decreased or Remained Unchanged[a]

Systolic Blood Pressure	Experimental No.	%	Control No.	%
Increased	20	8.2	4	2
Decreased	17	6.9	13	6.6
Unchanged	208	84.9	180	91.4
Total	245		197	

[a] Chi square = 8.19; degrees of freedom = 2; $p \le 0.02$.

Table 10. Number of Hypertensive Patients Whose Diastolic Blood Pressure Increased, Decreased or Remained Unchanged[a]

Diastolic Blood Pressure	Experimental No.	%	Control No.	%
Increased	7	2.9	0	
Decreased	9	3.7	8	4.1
Unchanged	229	93.4	189	95.9
Total	245		197	

[a] Chi square = 5.74; degrees of freedom = 2; $p \le 0.10$; many statisticians would consider this chi square to be nonsignificant since the expected frequencies in row one are less than five.

Table 11. Number of Diabetic Patients Whose Mean Fasting Blood Sugar Levels Increased, Decreased or Remained Unchanged[a]

Mean Fasting Blood Sugar Level	Experimental No.	%	Control No.	%
Decreased	54	29	27	20.9
Increased	35	18.8	36	27.9
Unchanged	97	52.2	66	51.2
Total	186		129	

[a] Chi square = 4.75; degrees of freedom = 2; $p \le 0.1$.

in patients with diabetes mellitus, as judged by fasting blood sugar concentrations. However, some significant differences were found in the kept-clinic-appointment rates, the frequency of clinic visits and the management of hypertensive patients.

The finding of a higher kept-clinic-appointment rate in the experimental group could be interpreted as a positive influence on patient compliance resulting from pharmacist intervention. The fact that the experimental group not only achieved a higher kept-appointment rate but also experienced a lower clinic dropout rate suggests that the clinical pharmacist was able to increase patient satisfaction with health care services as well.

In terms of patient use of health care services, one difference between the two groups was found. Patients receiving care from the clinical pharmacist were seen more frequently in the follow-up clinic than the patients cared for by the physician. The number of emergency room walk-in visits, referral clinic visits and hospital admissions were the same for both groups. The increase in compliance with clinic appointments could explain why the experimental patients averaged more follow-up clinic visits than the control patients.

The difference observed in the outcomes of care in the hypertensive patients is derived from a significantly lower mean systolic blood pressure in the patients who received care from the physician. It is important to note, however, that the mean diastolic blood pressures at the end of the trial were the same for both groups. One possible explanation for the higher mean systolic blood pressures found in the experimental group is that there were significantly more hypertensive patients in this group who also had diabetes mellitus. It is widely known that the vascular complications which can occur in diabetes may contribute to an increased systolic blood pressure. This was an uncontrolled variable in our experiment.

Several features associated with the design and execution of this study potentially may have biased the results. First, careful consideration should be given to the fact that all patient-care assessments and plans made by the clinical pharmacist were subsequently reviewed by physician auditors as a means of assuring provision of adequate medical care to the experimental patients. It should be pointed out, however, that the physician auditors rarely altered a plan written by the pharmacist. A prospective evaluation of physician acceptance of the pharmacist's therapeutic plans during the first 18 months of the study revealed that more than 99% of the plans were accepted without alteration.

Secondly, the method we used to measure the rate of compliance with antihypertensive drug regimens contains an obvious source of potential error: we had no proof that patients who obtained an appropriate number of drug refills actually took their medication as prescribed.

A third methodologic consideration is the way in which we calculated a kept-appointment rate for the control group. It was necessary to include the appointment data obtained from the residual group of patients, who also were receiving care from the physician. Thus, one must view the results pertaining to this particular criterion in terms of comparing an experimental group with an "expanded control group."

And, finally, the criteria we chose to measure outcomes of care may have been too insensitive to discern significant differences between the two groups of patients. The difference in the incidence of diabetic complications, for example, could have been significant despite comparable mean fasting blood sugar results. It also is conceivable that the treatment methods used in managing the hypertensive and diabetic patients could have greatly influenced the results in the outcomes of care. In that regard, an analysis of treatment methods (Table 12) revealed no difference in the type of treatment employed in the management of hypertension. However, the use of oral hypoglycemic agents and insulin in the treatment of diabetic patients was significantly different between the two groups.

Table 12. Method of Treatment of Diabetes Mellitus and Hypertension

Therapy	Experimental No.	Experimental %	Control No.	Control %	Significance
Diabetes mellitus	N = 186		N = 129		
Insulin	72	39	35	27	$p \leq 0.02$ ($z = 2.14$)
Oral hypoglycemic	101	54	88	68	$p \leq 0.01$ ($z = -2.40$)
Diet alone	13	7	6	5	NS
Hypertension	N = 245		N = 197		
Diuretic	232	95	190	96	NS
Sympathetic inhibitor	108	44	79	40	NS
Vasodilator	17	7	15	8	NS

[a] NS = no significant difference.

Conclusion

These potential pitfalls notwithstanding, we believe the research reported here adds further evidence to the safe and effective use of the clinical pharmacist in the postdiagnostic management of patients with chronic diseases. More work is needed to determine the cost-benefit relationship of this relatively new innovation in primary health care.

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